

REPTILES

Reptiles occupy a pivotal position in the history of vertebrate life—they were the first vertebrates to truly conquer the world's driest places. In their heyday, they evolved into dinosaurs—the biggest animals that have ever lived—but they diversified into birds and mammals as well. Although dinosaurs may have been

warm-blooded, all modern reptiles rely more on the warmth derived from the sun than from their metabolism to stay active. Their scaly skin—largely impervious to moisture—helps many to thrive in deserts, but thousands of species also live in wet, humid rainforests.

Anatomy

The archetypal reptile is a lizard—an elongated, scaly-skinned, long-tailed vertebrate that crawls on land on all four limbs, its body hanging close to the ground. The scales erupt from the outer layer of its skin and are reinforced with a tough protein called keratin—the same substance found in hair, feathers, and finger-nails. Reptiles replace their old scales either by shedding their outer skin in flakes or by sloughing it in one piece. In some kinds of lizards, the hindlimbs—or both sets of limbs—have been lost. All snakes lack limbs, but some primitive constrictors retain clawlike rudiments of the hindlegs. Turtles and tortoises have toothless beaks, but most reptiles have teeth—conical, needlelike, or jagged, but never differentiated into molars and canines as in mammals. The teeth of many snakes are modified as fangs for injecting venom.

Some lizards have particularly big scales, so they look armor-plated, but reptilian armor is especially solid in the bony shells of turtles and tortoises. The shells are usually covered with horny plates, but some are soft and leathery. Their upper shell is fused to the ribs in such a way that these reptiles have to breathe by pumping the muscles at the top of their legs. Crocodilians, the largest living reptiles, are armored differently: they have plates—called osteoderms—that are reinforced with bone.

Reproduction

All reptiles reproduce by internal fertilization that involves copulation. Unlike the soft-coated eggs of amphibians, reptiles produce eggs that are surrounded by a protective membrane, called an amnion, derived from the developing embryo.

Reptiles share this feature with birds and mammals, so these vertebrates are collectively called amniotes. The evolution of the amnion was critical in allowing vertebrates to survive on dry land. Reptile eggs have a leathery or chalky shell. A few lizards and snakes retain the developing eggs in their reproductive system and give birth to live young. Many reptilian parents guard their eggs or young, and a few build nests. Crocodilians show special care, and when the eggs hatch, the mother carries the hatchlings to a nearby pool for release.

Behavior

Although tortoises and some lizards are herbivorous, most reptiles are predatory. Reptiles have well-developed senses for tracking their prey on dry land. The sense of smell is especially well developed—most lizards and all snakes have a forked tongue that collects

▽ SENSING DIRECTION

Typical of other snakes—and most lizards—the mangrove snake's forked tongue enables it to sense scent molecules coming from different directions, helping it to track its prey.



unusually long vertebral column

△ SKELETON WITHOUT LIMBS

Reptiles have a bony skeleton, which is highly modified in limbless snakes. Pythons and related constrictors, however, retain claw-like remnants of the pelvic girdle.



△ BREAKING OUT

Although some reptiles—such as sea turtles—have conquered the oceans, their eggs must hatch—and hence, are laid—on land. Only live-bearing sea snakes are permanently aquatic.

▷ PROTECTIVE COVERING

Waterproof skin lets reptiles cope with some of the driest habitats. Some reptiles, such as this agama lizard, use colorful skin to make territorial claims.

scent molecules and transfers them to a sensory organ, called Jacobson's organ, in the roof of the mouth. Some snakes also have sensory pits on their head that detect the infrared radiation given out by warm-blooded prey.

Most reptiles have a sluggish metabolism compared to mammals and birds, so they generate limited body warmth. They rely on the environment to raise their body temperature to a level at which they can become active. Most achieve this by basking in the sun.

REPTILE CLASSIFICATION

There are four orders of reptiles. Most species belong to one of these orders, the squamates, which includes lizards, snakes, and amphisbaenians.

Tortoises and turtles » p428
Order Testudines Species 341

Tuatara » p358
Order Rhynchocephalia Species 1

Snakes and lizards » pp429-35
Order Squamata Species c.9,810

Crocodiles and alligators » p435
Order Crocodilia Species 25